Venkata Rami Reddy Bobbala

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Summary

Java Backend Developer with 4 years of experience architecting and delivering scalable, production-grade distributed systems using Java and the Spring ecosystem. Proven track record at Tata Consultancy Services building event-driven microservices with Kafka, low-latency caching layers with Redis, and containerized deployments via Docker. Cut boilerplate by 25%, improved API throughput by 40%, and consistently shipped features in fast-paced Agile environments. Adept at database performance optimization, JWT-secured API design, and maintainable software architecture.

Technical Skills

Languages: Java (8 / 11 / 17), SQL
Frameworks & Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring JDBC, Spring Security (JWT), Spring AOP, Spring AI, REST APIs, Microservices Architecture
Messaging & Caching: Apache Kafka, Redis
Databases: PostgreSQL, MySQL
DevOps & Tooling: Docker, Git, Maven, JUnit, Mockito, CI/CD Pipelines, Linux

Professional Experience

Tata Consultancy Services (TCS) Hyderabad, India
Java Backend Developer *Dec 2021 – Dec 2025*

- Architected and maintained scalable backend microservices using **Java 17**, **Spring Boot**, and **Spring MVC** for large-scale enterprise applications, supporting **10,000+ concurrent users** with 99.9% uptime.
- Designed and integrated **Apache Kafka** event-driven pipelines for asynchronous inter-service communication, reducing service coupling and improving system throughput by **40%** under peak load.
- Implemented **Redis** distributed caching using Spring Cache and Lettuce, cutting average API response times by **60%** for high-frequency read operations and eliminating redundant database hits.
- Containerized microservices using **Docker**, authoring Dockerfiles and Compose configurations that standardized dev/prod environments and reduced deployment setup time by **30%**.
- Delivered backend REST APIs end-to-end – from schema design and service logic through integration testing and production deployment – across multiple enterprise feature releases.
- Drove database performance improvements across **PostgreSQL** and **MySQL** by applying strategic indexing, query refactoring, and efficient data-access patterns via Spring JPA and Spring JDBC, reducing slow query count by **35%**.
- Engineered **JWT-based authentication** and role-based authorization using **Spring Security**, hardening API endpoints against unauthorized access across production services.
- Led production incident investigations, diagnosing root causes through log analysis and performance profiling, delivering hotfixes with minimal service disruption and **sub-2-hour MTTR**.

Projects

AI-Powered Enterprise Search Service | ***Spring AI, Spring Boot, Spring Security, Spring Data JPA*** GitHub

- Built a production-grade AI-powered semantic search backend using **Spring AI** and vector database integration, enabling accurate enterprise document retrieval far beyond traditional keyword matching.
- Implemented document embedding pipelines and semantic similarity scoring, significantly improving search relevance and reducing time-to-result for internal knowledge queries.
- Secured all API surfaces with **JWT authentication** and role-based access control (RBAC) via Spring Security, ensuring strict data isolation between user roles.
- Leveraged Spring Data JPA for efficient metadata persistence and retrieval, supporting complex filtering and pagination across large document corpora.
- Designed a modular, extensible backend architecture with clear abstraction layers, providing a ready foundation for **Retrieval-Augmented Generation (RAG)** capabilities in future iterations.

Education

ACE Engineering College Hyderabad, India
Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering *2017 – 2021*